OEE Targets and Escalation Matrix

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> **Audience:** Plant Manager · Line Manager · Corp Exec · Maintenance Lead **Applies to:** All 5 production lines.

1. Purpose

This document is the corporate KPI definition for **Overall Equipment Effectiveness (OEE)** at this site, the **target values** each line is held to, and the **escalation matrix** that fires when targets are missed.

2. OEE definition (matches the Fabric Dashboard)

```
OEE = Availability × Performance × Quality
```

- **Availability** = `Running` time ÷ (`Running` + `Fault` + `Maintenance`) time. `Idle` time (`Starved` or `Blocked`) is **excluded** because it reflects neighboring-station issues, not the station's own availability.
- **Performance** = ideal cycle time ÷ actual cycle time.
- **Quality** = (parts produced − rejected parts) ÷ parts produced.

The Fabric Eventhouse `OEE_5min` materialized view computes these components on rolling 5-minute windows by station and line. The Real-Time Dashboard surfaces them on the Plant Manager and Line Manager pages.

3. Site-level targets

Tier	Target OEE	Comment
World-class	≥ 85 %	Aspirational; achievable only on Line-B with bottleneck balanced.
This site's commitment	**≥ 75 %**	Quarterly review with Corp Exec.
Acceptable	60 – 75 %	Normal operating band.
Action required	< 60 %	Escalation triggers fire — see §5.

4. Line-level targets

Line	Bottleneck	Target OEE	Notes
Line-A	CMM-Inspection (60 s)	78 %	Quality target capped by CMM reject rate (≤ 5 %).
Line-B	Hydraulic-Press fault rate	72 %	Availability target capped by 1.5 % fault rate.
Line-C	Welding-Robot + Leak-Test	75 %	Watch reject rate from Weld-Inspection.
Line-D	Curing-Oven (90 s)	70 %	Slowest line; Performance capped by oven cycle.
Line-E	Wave-Solder fault rate	73 %	Highest fault-rate line; quality from Functional-Test.

5. Escalation matrix

Condition	Sustained for	Notify	Action
Station OEE < 60 %	> 15 min	Line Manager	Pause for triage; check statio
Line OEE < 60 %	> 15 min	Plant Manager	Open incident review; susper urgent changes.
Line OEE < 60 %	> 1 hr	Corp Exec	Daily-flash report.
≥ 3 faults on the same station	within 1 hr	Maintenance Lead	Pull station for PM regardles schedule; root-cause review.
Reject rate > 2× target	within 1 hr	Quality Engineer	Quarantine all buffer parts or station per `Quality_Reject_Disposition_P`
Two concurrent open work orders on the same line	any time	Line Manager + Maintenance Lead	Re-balance technicians; chec `Maintenance_Workflow_Overvie`
Bottleneck station downtime (CMM, Curing-Oven, Wave-Solder)	> 5 min	Line Manager	Notify Plant Manager; check SOP.

6. Bottlenecks and OEE sensitivity

A 5-minute outage on a bottleneck station translates almost directly to 5 minutes of lost line throughput. Watch carefully:

- **Line-A CMM-Inspection** — see `Line-A_05_CMM-Inspection_Calibration.pdf`.
- **Line-D Curing-Oven** — see `Line-D_05_Curing-Oven_SOP.pdf`.
- **Line-E Wave-Solder** — see `Line-E_06_Wave-Solder_Troubleshooting.pdf`.

Non-bottleneck stations can absorb short outages within buffer capacity (5 parts each), but cascading starvation through the line still erodes OEE within ~1–2 cycle times of the slowest station.

7. References

- `Maintenance_Workflow_Overview.pdf` — fault → WO → resolution.
- `Quality_Reject_Disposition_Policy.pdf` — Quality component handling.
- `Safety_Lockout_Tagout_Procedure.pdf` — all maintenance work requires LOTO.
- `Production_Lines_Reference.pdf` — line topology.
- `Shift_Handover_Checklist.pdf` — OEE component is reported on handover.